Microbiology: Bacterial Respiratory Pathogens I

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Session Objectives

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1. List and differentiate bacterial infections of the respiratory system associated with community acquired pneumonia.
2. Describe the unique microbiologic features and any associated virulence factors for each pathogen
3. Explain any relevant symptoms and diagnostic tests for each pathogen.
4. Describe any treatment or prevention strategy for each pathogen.

Types of Respiratory Infections

- Upper Respiratory Tract:
  - Sinusitis
  - Otitis media
  - Otitis externa
  - Pharyngitis
  - Epiglottitis
- Lower Respiratory Tract:
  - Bronchitis
  - Pneumonia

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Pneumonia

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- Inflammation of the lung affecting the alveoli
  - Alveoli can become filled with fluid or pus
- Epidemiology:
  - Over 4.5 million cases per year in USA
  - Most common infectious cause of death
- Pneumonia is frequently categorized based on site of acquisition
  - Community acquired (both typical and atypical)
  - Nosocomial pneumonia
    - hospital-acquired pneumonia & ventilator-associated pneumonia
- Diagnosed by chest x-ray: “shadow” or opaque area is present over a portion of the lung
- Age and predisposing factors are important for diagnosis

Pneumonia Causes by Group

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- Alcohol overuse: *Klebsiella*, anaerobes due to aspiration
- IV Drug use: *S. pneumoniae, S. aureus*
- Cystic fibrosis: *Pseudomonas*, *S. aureus, S. pneumoniae*, *Burkholderia cepacia*
- Immunocompromised: Fungi, *P. jirovecii* (with HIV)
- COPD: *S. pneumoniae, H. influenzae, M. catarrhalis, Pseudomonas*

Important Bacterial and Viral Causes of Community-Acquired Pneumonia by Age (Listed in Order of Frequency)

Age	Bacteria	Viruses
Neonates	Group B streptococci *Escherichia coli*	Respiratory syncytial virus (RSV)
Infants	*Chlamydia trachomatis* *Streptococcus pneumoniae*	RSV Parainfluenza virus
Children	*S. pneumoniae* *Haemophilus influenzae*	RSV Parainfluenza virus
Young adults	*Mycoplasma pneumoniae* *Chlamydophila pneumoniae* *S. pneumoniae*	Various respiratory viruses (e.g., adenovirus)
Older adults	*S. pneumoniae* *H. influenzae* *Legionella pneumophila*	Influenza virus SARS-CoV-2[1]

Bacterial Species

- Community-acquired pneumonia
  - *Streptococcus pneumoniae*
  - *Streptococcus pyogenes*
  - *Staphylococcus aureus*
  - *Haemophilus influenzae*
  - *Klebsiella pneumoniae*
  - *Bordetella pertussis*
- Atypical pneumonia
  - *Mycoplasma pneumoniae*
  - *Legionella pneumophila*
  - *Chlamydophila (Chlamydia) pneumoniae*

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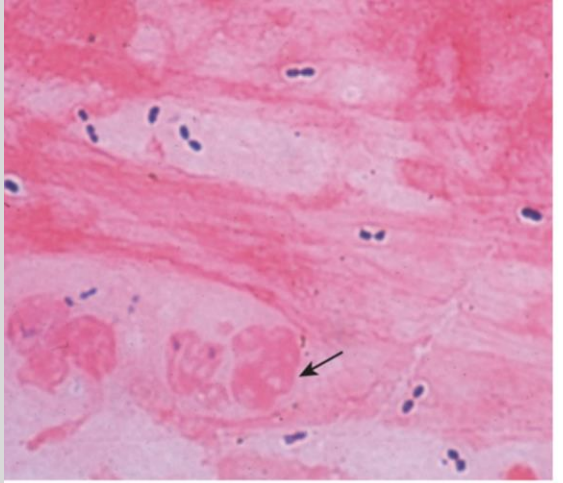
Community-acquired pneumonia

- ***Streptococcus pneumoniae***
- ***Streptococcus pyogenes***
- ***Staphylococcus aureus***
- ***Haemophilus influenzae***
- ***Klebsiella pneumoniae***
- ***Bordetella pertussis***

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Streptococcus pneumoniae

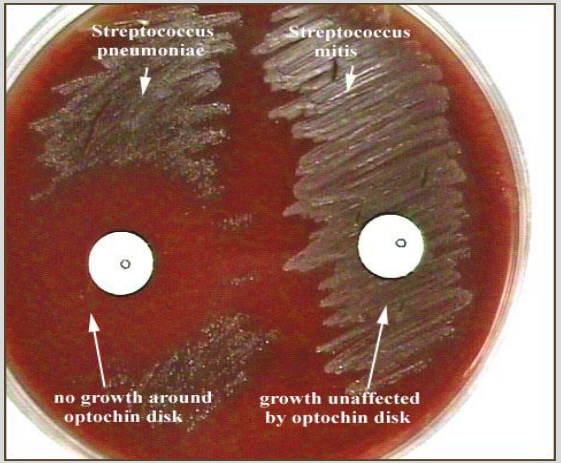
- Characteristics:
  - **Gram +ve**
  - Lancet-shaped diplococci/chains
  - Catalase –ve
  - α-hemolytic
  - Optochin sensitive and bile soluble
- Urine antigen test
- Virulence:
  - IgA protease
  - Encapsulated
    - Basis for vaccines



Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchell J.A. Sakanari, P. Hotez, R. Mejia Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e
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Most commonly causes **MOPS**:

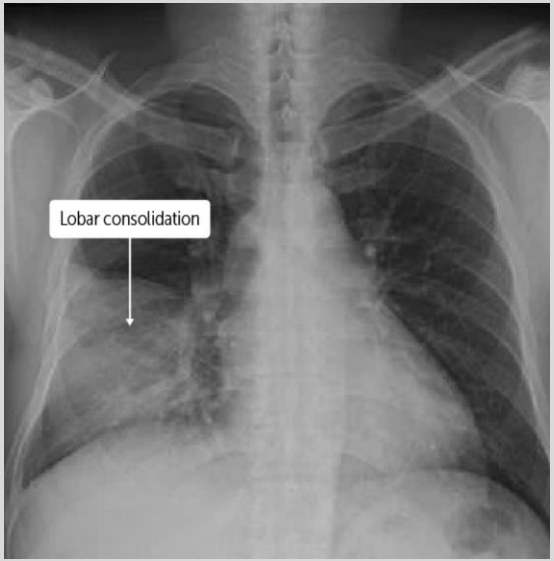
Meningitis
Otitis media
Pneumonia
Sinusitis



Streptococcus pneumoniae

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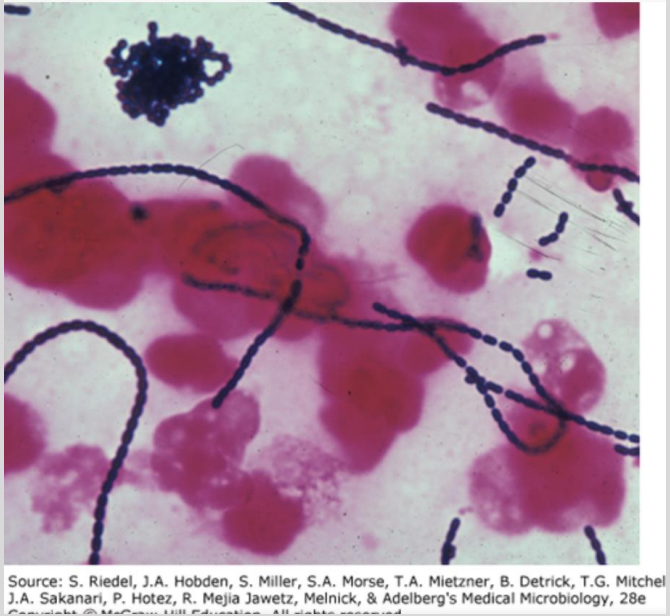
- Pneumococcal pneumonia
  - S/sx: sudden onset with fever, chest pain & thick, rusty sputum
  - Common in young-to-middle-age adults- rare but lethal in infants and elderly
  - Infection: Aerosolized bacteria are inhaled into alveoli or already colonized bacteria are aspirated into lungs
  - Classic lobar form
  - Asplenic patients are predisposed



Streptococcus pyogenes

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- Aka Group A Strep (GAS)
- Characteristics:
  - Gram +ve cocci in chains
  - Bacitracin sensitive
  - β-hemolytic
  - Catalase -ve
- Virulence:
  - Streptokinase, streptolysin S & O, DNase; Capsule contains hyaluronic acid & “M” protein (found in cell membrane & wall) and these lead to resistance to phagocytosis



Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchel J.A. Sakanari, P. Hotez, R. Mejia Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e

Streptococcus pyogenes

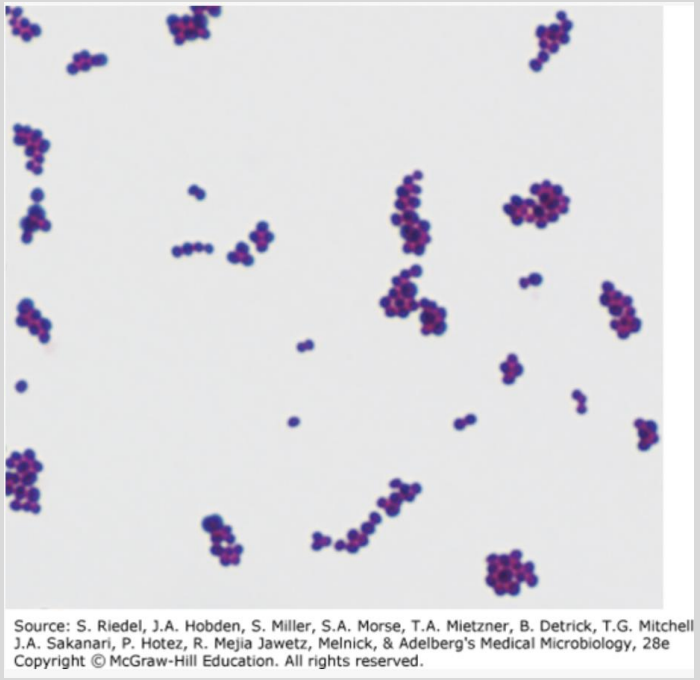
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- Transmission: respiratory droplets
- Causes
  - strep pharyngitis (which can lead to rhematic fever and glomerulonephritis), and rarely pneumonia
  - Also causes cellulitis, impetigo, erysipelas, scarlet fever
- Dx: clinical dx; rapid strep antigen test and/or culture swab on blood agar; look for hemolysis and bacitracin sensitivity; also can detect antibodies for prior infection (ASO titers)

Staphylococcus aureus

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- Characteristics:
  - Gram +ve
  - Cocci in clusters
  - β-hemolytic
  - Catalase +ve
  - Coagulase +ve
- Virulence:
  - Protein A binds Fc-IgG → inhibits complement activation and phagocytosis



Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchell J.A. Sakanari, P. Hotez, R. Mejia Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e
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Haemophilus influenzae

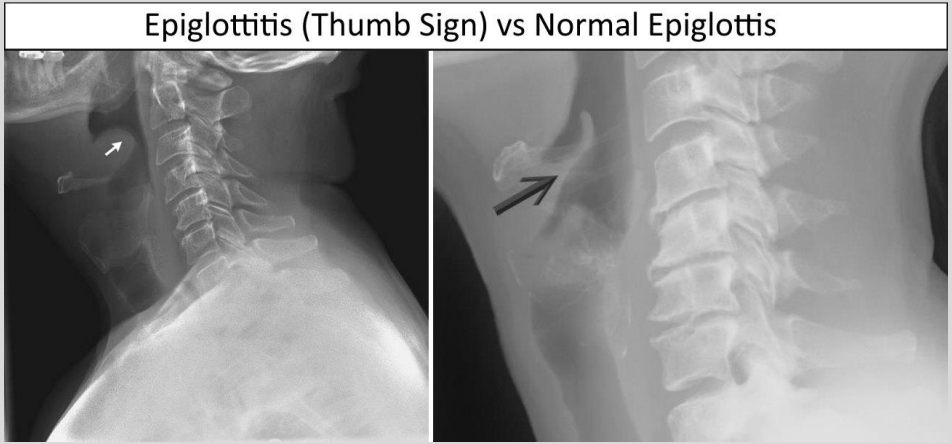


- Aka *H. flu*
- Characteristics:
  - **Gram –ve** coccobacillus
  - Growth on chocolate agar (factors V (NAD) and X (hemin))
- Virulence:
  - IgA protease; Encapsulated: polyribosyl ribitol phosphate (PRP)
  - Six capsular antigenic types: a, **b**, c, d, e, f
  - Basis for Hib vaccine
- Transmission: droplets

Haemophilus influenzae

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- *Haemophilus* causes:
- Epiglottitis
  - unvaccinated children
  - "cherry red" appearance of epiglottis on endoscopy and "thumb sign" on lateral neck X-ray
- Meningitis
- Otitis media
- Pneumonia

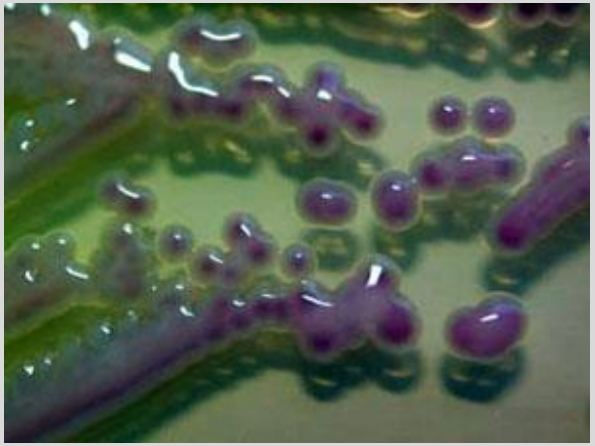


https://www.grepmed.com/images/10109/comparison-lateral-neckxray-normal-epiglottis-57

Klebsiella pneumoniae

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- Characteristics:
  - **Gram –ve** Bacillus
  - Enteric, Non-motile, Lactose-fermenter
- Can be aspirated from GI tract and then reach respiratory tract
- Virulence:
  - Endotoxin (or LPS)
  - Polysaccharide capsule
- Causes lobar pneumonia, commonly in patients with alcohol abuse
- S/sx: fever, dyspnea “currant jelly” sputum



ABCDE’s of Klebsiella:
Aspiration pneumonia
aBscess in lungs and liver
Currant jelly sputum
Diabetes mellitus
EtOH overuse

Bordetella pertussis

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- Characteristics:
  - Gram –ve aerobic coccobacillus
  - Grows only on Bordet-Gengou selective media
- Virulence:
  - Pertussis Toxin: Leads to edema in respiratory mucosa; Produces lymphocytosis
  - Adenylate Cyclase Toxin: Directly impairs phagocytes
  - Tracheal Cytotoxin: Kills ciliated epithelial cells
- Three clinical stages of whooping cough:
  - Catarrhal, Paroxysmal- intense cough followed by inspiratory “whoop”, Convalescent
- Prevention: Tdap and DTaP vaccines

Atypical pneumonia

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- Common bacterial species

- *Mycoplasma pneumoniae*
- *Chlamydophila pneumoniae*
- *Legionella pneumophila*

Mycoplasma pneumoniae

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- Characteristics:
  - NO CELL WALL: Pleomorphic, no Gram stain reaction
  - Sterols in membrane for stability
  - Grows on Eaton agar
- Causes “walking pneumonia”
- Frequently in those <30 y/o
- Outbreaks in military recruits, prisons, colleges, etc.

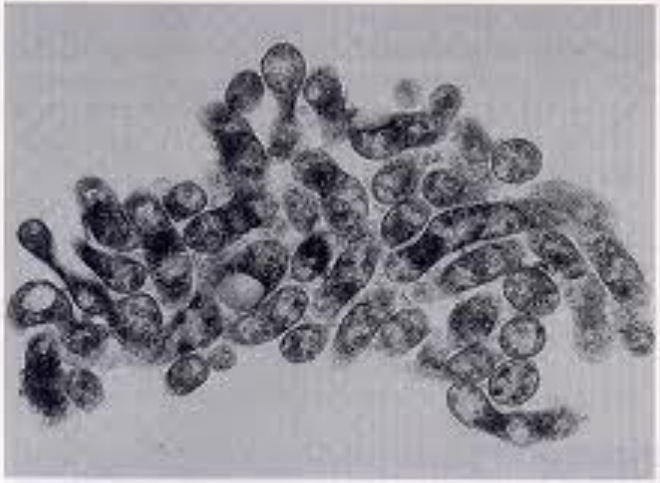
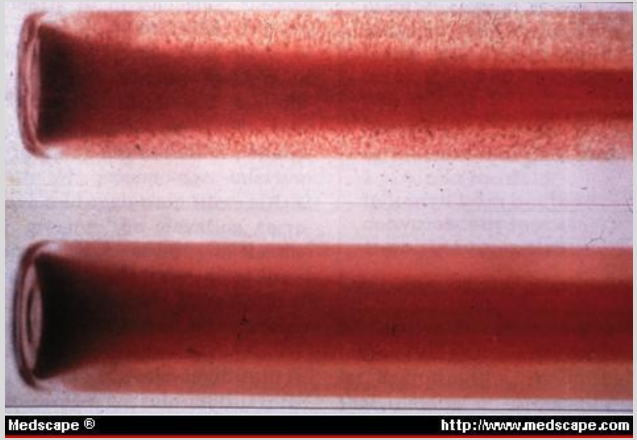


Fig. 17-83 ***Mycoplasma.*** Electron micrograph of *Mycoplasma pneumoniae*. The cell lacks a cell wall and is bounded by a cytoplasmic membrane that has a trilaminar structure.

Mycoplasma pneumoniae

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- s/sx: nonproductive cough, fever, headache, malaise, etc.
- Dx:
  - Clinical; chest X-ray shows infiltrate
  - Serology (cold agglutinins)
    - IgM antibodies cross-react with RBCs at cold temperatures → agglutination
- Tx: macrolides, doxycycline, or FQ's (non-beta lactam abx because no cell wall!)



Chlamydophila pneumoniae

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- Characteristics:
  - Obligate intracellular organism
  - Cell wall lacks extensive peptidoglycan due to decreased muramic acid
  - 2 forms:
    - Elementary body: small, dense, infectious, and enters cells via endocytosis
    - Reticulate body: replicates in cell; develop into elementary bodies
- Transmission: aerosol
- Lab Dx:
  - PCR or other NAATs; Serology; reticulate bodies seen on Giemsa stain
- Tx: Doxycycline or Macrolides

Legionella pneumophila

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- Characteristics:
  - Gram –ve rod (stains poorly), silver stain (biopsy)
  - Grows on BCYE medium with iron and cysteine
- Transmission:
  - Aerosol from environmental water source
  - No person-to person transmission
- Legionnaires' disease:
  - Mild flu-like to severe pneumonia (with possible GI and CNS symptoms)
  - Related to Pontiac fever
- Diagnosis:
  - Detected by presence of antigen in urine
  - Labs show hyponatremia
- Tx: macrolide or quinolone

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Questions for review- answers in notes section of PPT

Practice Question #1

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Q: A 65-year-old man with chronic obstructive pulmonary disease presents to clinic with sudden onset fever, productive cough with rust-colored sputum, and pleuritic chest pain. Chest X-ray shows lobar consolidation. Which virulence factor of the most likely pathogen allows it to evade phagocytosis?

A. IgA protease
B. Polysaccharide capsule
C. M protein
D. Exotoxin A
E. Lipooligosaccharide

Practice Question #2

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Q: A 7-year-old child presents to clinic with fever, sore throat, and tender anterior cervical lymphadenopathy. On examination, the tonsils are erythematous with exudates. Rapid strep test is positive. Which complication is most closely associated with untreated infection by this pathogen?

A. Guillain-Barré syndrome
B. Rheumatic fever
C. Toxic shock syndrome
D. Scarlet fever

Practice Question #3

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Q: A 62-year-old man presents with fever, cough, watery diarrhea, and confusion. He recently attended a convention at a hotel. Labs show Na^+ = 126 mEq/L. CXR reveals patchy interstitial infiltrates. Which of the following is the most likely causative organism?

A. *Legionella pneumophila*
B. Mycoplasma pneumoniae
C. Streptococcus pneumoniae
D. Chlamydophila pneumoniae

Summary Slide

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- Community-acquired pneumonia
  - *Streptococcus pneumoniae:* Gram + lancet diplococci; α-hemolytic; Catalase -; optochin sensitive; bile soluble; encapsulated
  - *Streptococcus pyogenes:* Gram + cocci chains; β-hemolytic; catalase -; bacitracin sensitive; ASO titers; hyaluronic acid and M protein virulence; strep pharyngitis
  - *Staphylococcus aureus:* Gram + cocci in clusters; β-hemolytic; catalase +; coagulase +; Prtn A
  - *Haemophilus influenzae:* Gram – rod; chocolate agar; epiglottitis
  - *Klebsiella pneumoniae:* Gram – rod; lactose fermenter on MacConkey agar; enteric; encapsulated; currant jelly sputum; alcohol use, diabetes, aspiration
  - *Bordetella pertussis:* Gram – coccobacillus; Bordet-Gengou media; pertussis toxin; 3 clinical stages
- Atypical pneumonia
  - *Mycoplasma pneumoniae:* sterol membrane; Eaton agar; cold agglutinins
  - *Legionella pneumophila:* Gram – rod (stains poorly); charcoal yeast extract medium; hyponatremia on labs; heating/cooling systems; Legionnaire's disease
  - *Chlamydophila pneumoniae:* obligate intracellular; Elementary body vs reticulate body

References

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THANK YOU!

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